TED ZHAO

416 551 622 · zp4work+job@gmail.com
10/4 Sheppards Lane, Norwood SA 5067, Australia
linkedin.com/in/ted-zhao · github.com/tade0726

PROFILE

- Over 5 years of experience in Machine Learning and Data Science, specializing in recommendation systems and computer vision.
- Experienced in delivering end-to-end ML solutions using Databricks/Spark, Docker, and Python.
- Skilled in conducting AB tests and tracking metrics for real-world business applications.
- Experienced in Python backend development, with familiarity in Django, Flask, and SQLAlchemy, for side projects and additional responsibilities.

SKILLS

Programming Languages	**Python, VBA, Scala, Julia**
Web/Web Scraping	Django, Flask, asyncio, Scrapy
AI/Machine Learning	Reinforcement Learning, Deep Learning, Clustering, Lightgbm, XGBoost, SVM, Word2Vec
Data Science	Tableau, A/B Testing, Statistics, Data Analysis
ETL	SQL, SQLAlchemy, Spark, Pandas
Cloud	AWS, AWS Sagemaker, Azure ML, Databricks
Dev	Git, Docker

WORK EXPERIENCE

Machine Learning Engineer - Skyscanner

Shenzhen, China (Collaborated with teams across the UK and Bulgaria) *Nov 2019 - July 2021*

Recommendation Systems

- Developed and improved recommendation systems using iterative ML models (Logistic Regression / LightGBM / Deep learning).
- Utilized **Docker** for end-to-end model delivery with **AWS Sagemaker** environment, resulting in a **+5-10% increase in CTR** with each iteration.

Computer Vision / Deep Learning

- Applied Deep learning models to hotel images to enhance user experience.
- Employed ResNet for categorizing hotel images into 20 landscape/interior labels, achieving **99% accuracy** in Top-5 predictions.
- Utilized MobileNet to select thumbnails for hotels, resulting in a **+3% increase in CTR**.

A/B Testing

- Performed AB test analysis for car-hire and hotel search pages.
- Achieved **10X growth** in car-hire during the pandemic.
- Built data dashboards in Tableau and Databricks for stakeholder reference.

ETL

- Designed and implemented data pipelines in Spark/Databricks.
- Generated **a dozen key metrics and features** to train models and conduct AB testing analytics.

Data Scientist - Smart Decision Technology Ltd

Shenzhen, China *Aug 2015 - Mar 2018*

Simulation

- Designed and implemented a Python-based simulation system for S.F. Express.
- Leveraged **SimPy** to accurately model the conveyor system within the Distribution Centre.
- Created an intuitive front-end interface, enabling business users to input site designs and parameters with ease.
- Developed a robust back-end service using **Flask**, which stored simulation parameters and results for each iteration, utilizing **SQLAlchemy** for efficient model management.
- The solution surpassed three competitors in terms of speed and adaptability, securing a **2-year contract** with S.F. Express.

Credit Risk Modeling

- Designed credit scorecard workflows using Python to replace SAS.
- Created **feature selection, binning, and WOE calculation** in Python.
- Converted the logistics model into scorecards in Python.

Segment Analysis / NLP

- Delivered client segment reports for Hong Kong Jockey Club.
- Created segment analysis using clustering and generated data diagrams and charts in reports.
- Built **sentiment analysis using Word2Vec** on mixed Cantonese and English restaurant reviews.

Data Scientist - DJI

Shenzhen, China *April 2018 - Aug 2018*

Competitive Analysis

- Provided support for a comprehensive campaign through competitive analysis.
- Scraped retail location data from competitors with **Scrapy**.

Data Reports

- Analyzed collected dataset to identify patterns and insights for the campaign team.
- Translated findings into actionable recommendations.
- Empowered stakeholders to make informed decisions and optimize the product launch strategy.

Django

- Developed a Django-based API for Jupyter integration extension, enabling seamless SQL querying and data model management.
- Designed the API to facilitate Jupyter extension integration for streamlined SQL query execution.
- Maintained data models using Django and SQLAlchemy, ensuring standardized datasets across the platform.

RESEARCH PROJECT

Quantitative Trading with Reinforcement Learning - The University of Adelaide

Adelaide, SA, Australia *Sep 2022 - Present*

Reinforcement Learning

- Conducting a research project while pursuing a master's degree in Machine Learning at The University of Adelaide.
- Developed a trading strategy for the Crypto market using **Reinforcement Learning**.
- Designed a reward system with Sharp Ratio, where actions were the next sell/buy quantity of all assets.
- Created **two Deep Learning modules** to capture price signals and market sentiment.

Back Testing

- Developed a back-test with metrics **Sharp Ratio, Maximum Drawdown(MDD)** and presented profit curves with plots, considering fees and compound returns.

Education

	Institution	Duration
Master of Machine Learning	The University of Adelaide	June 2021 - June 2023
Bachelor's Degree in Mathematical Sciences	Shenzhen University	September 2010 - July 2014

Certifications

	Date	Verification
Machine Learning - Coursera	October 2016	Verify
Machine Learning Nanodegree Program - Udacity	December 2018	Verify

Language Proficiency

	Proficiency
English	IELTS - CEFR C1 (Advanced)
Mandarin (Chinese)	Native
Cantonese (Chinese)	Native